

Assignment 2: Financial Engineering

Group 12

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1. Case study: Interest Rate Curve Bootstrap

The discount curve $B(t_0, T)$ is built sequentially from three instrument classes: deposit rates, STIR futures, and swap rates. All instruments share the settlement date $t_0 = 15$ February 2008.

1. Deposits

A deposit with maturity T_i and rate L_i (act/360 day count) gives

$$B(t_0, T_i) = \frac{1}{1 + \delta_i^{\text{dep}} L_i}, \quad \delta_i^{\text{dep}} = \frac{T_i - t_0}{360}.$$

Only deposits whose end date falls *before* the expiry of the first futures contract are retained, so as to avoid overlapping instruments.

2. STIR futures

Each futures contract i spans the period $[t_i, t_{i+1}]$ and quotes a forward rate F_i (act/360). The strip is bootstrapped forward iteratively:

$$B(t_0, t_{i+1}) = B(t_0, t_i) \cdot \frac{1}{1 + \delta_i^{\text{fut}} F_i}, \quad \delta_i^{\text{fut}} = \frac{t_{i+1} - t_i}{360},$$

where $B(t_0, t_i)$ is recovered by log-linear (zero-rate) interpolation on the deposit pillars already bootstrapped:

$$B(t_0, t) = \exp(-z(t) \cdot t), \quad z(t) = \text{interp}(\{t_k\}, \{z_k\}, t), \quad z_k = -\frac{\ln B(t_0, T_k)}{(T_k - t_0)/365}.$$

3. Swaps

A vanilla IRS with tenor T_n , fixed rate S , and annual coupon dates $t_1 < \dots < t_n = T_n$ satisfies the condition

$$1 = S \sum_{k=1}^n \delta_k B(t_0, t_k) + B(t_0, T_n),$$

where $\delta_k = \text{yearfrac}(t_{k-1}, t_k)$ under the 30/360 European convention. Solving for the terminal discount factor:

$$B(t_0, T_n) = \frac{1 - S \text{BPV}_{n-1}}{1 + \delta_n S}, \quad \text{BPV}_{n-1} = \sum_{k=1}^{n-1} \delta_k B(t_0, t_k),$$

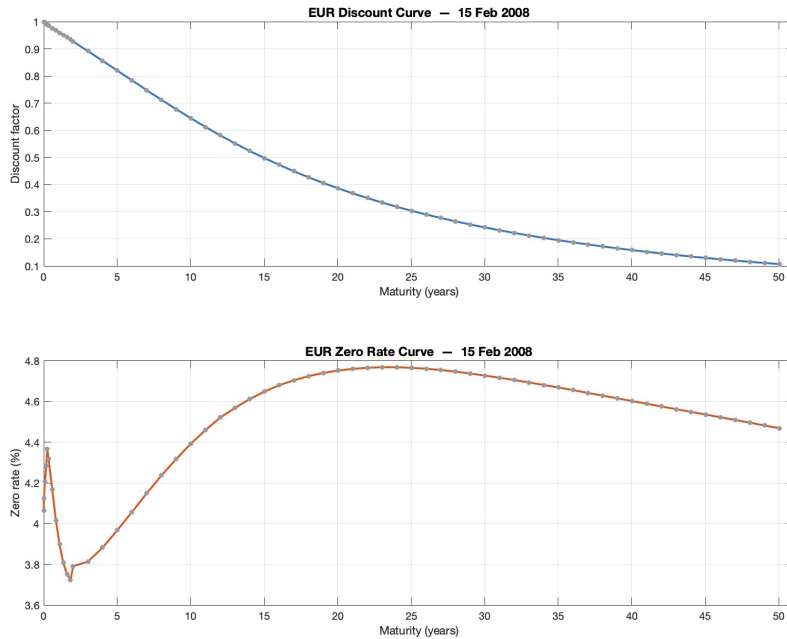
where intermediate discounts $B(t_0, t_k)$ for $k < n$ are obtained by log-linear zero-rate interpolation on the pillars already bootstrapped from deposits and futures. The procedure is applied sequentially for each swap maturity, extending the curve one pillar at a time.

4. Zero rates

Throughout the bootstrap, continuously compounded zero rates on the act/365 basis are maintained as working quantities:

$$z(T) = -\frac{\ln B(t_0, T)}{(T - t_0)/365},$$

and are updated after each new pillar is added. The final output consists of the triplet $\{T_i, B(t_0, T_i), z(T_i)\}$ for all bootstrapped maturities.



Q1: P&L Estimation and Quality Assessment

1. Theoretical Framework and Assumptions

The IRS PL at inception is simply the difference between the executed and fair rates, scaled by the Basis Point Value (BPV).

The general formula for the Net Present Value (NPV) of the swap, from the perspective of a fixed-rate receiver, is:

$$P\&L_{bank} = N \cdot (R_{trade} - R_{bank}) \cdot BPV_{mkt}$$

Where:

- N is the *Notional amount*, equal to 100,000,000 € .
- R_{trader} is the fixed rate received by the trader, equal to 4.117%.
- R_{bank} is the wrong rate.
- BPV is the *basis point value* of the fixed leg, defined as $BPV = \sum_{i=1}^n \tau_i \cdot DF(t_i)$, representing the sum of the discount factors multiplied by their respective year fractions.

Because Bank XX employs a nonstandard methodology to obtain its discount factor curve, we do not have access to their exact internal discount factors. Therefore, to perform our estimation, we pragmatically approximate the Basis Point Value using the market discount factors (BPV_{mkt}) obtained from a standard bootstrap.

2. P&L Estimation: Internal vs. Market Perspective

To fully comprehend the trade dynamics, we must estimate the P&L under two distinct scenarios.

A. The Bank's Internal Perspective (Mark-to-Model)

According to Bank XX's nonstandard discount curve, the fair 6-year swap rate vs 3M is estimated at 4.027%. Using this rate as our benchmark (R_{bank}):

$$\begin{aligned} P\&L_{bank} &\approx 100,000,000 \cdot (0.04117 - 0.04027) \cdot A_{mkt} \\ P\&L_{bank} &\approx 100,000,000 \cdot (0.00090) \cdot A_{mkt} = +\mathbf{90,000} \cdot \mathbf{A_{mkt}} \end{aligned}$$

Result: From the bank's internal perspective, the trader is receiving 90 basis points above the internally perceived fair rate.

B. The True Market Perspective (Mark-to-Market)

In reality, the liquid interbank market prices the 6-year mid-market swap rate at 4.127%. Using the actual market rate as our benchmark (R_{mkt}):

$$\begin{aligned} P\&L_{mkt} &= 100,000,000 \cdot (0.04117 - 0.04127) \cdot A_{mkt} \\ P\&L_{mkt} &= 100,000,000 \cdot (-0.00010) \cdot A_{mkt} = -\mathbf{10,000} \cdot \mathbf{A_{mkt}} \end{aligned}$$

Result: In real economic terms, the trader closed the deal at a rate 10 basis points lower than the market mid-rate. The trade is actually loss-making, as the trader essentially crossed the bid-ask spread and destroyed value for the bank.

3. Estimation Quality and Final Remarks

Regarding the quality of this estimation, a crucial distinction must be drawn between our mathematical approximation and the bank's pricing framework.

Our mathematical estimation, based on A_{mkt} , is of excellent quality and accurately captures the monetary magnitudes involved. However, the quality of Bank XX's internal pricing methodology is critically flawed. The nonstandard curve creates a dangerous optical illusion: it generates a highly positive accounting P&L (*mark-to-model*) while concealing a tangible economic loss (*mark-to-market*). The estimation is quite imprecise due to the fact that the methodology adopted by the bank is unknown and so that the

Exercise 2: I.B.Coupon Bond

The bond has face value $N = 100$ Mln, coupon rate $\bar{c}$ equal to the 8y par swap rate S , and the same reset and payment dates as the corresponding IRS. The price is determined without explicit discounting by decomposing the bond into swap legs.

Decomposition

The IB coupon bond can be decomposed as the *fixed leg of the IRS* — which generates the same coupon cash flows — augmented by two notional exchanges that are absent in the swap: the investor pays N at t_0 and receives N at maturity t_N . Formally:

$$P = \underbrace{\bar{c} N \sum_{i=1}^n \delta_i B(t_0, t_i)}_{\text{fixed coupons}} + \underbrace{N B(t_0, t_N)}_{\text{face value}}.$$

Since $\bar{c} = S$, the IRS has zero NPV at inception, so the fixed leg equals the floating leg:

$$\bar{c} N \sum_{i=1}^n \delta_i B(t_0, t_i) = N \sum_{i=1}^n E[\delta(t_{i-1}, t_i) L(t_{i-1}, t_i) B(t_0, t_i)].$$

The bond price therefore equals the price of an *IB floater*: a floating-rate bond with the same maturity and face value.

Floater valuation via telescopic sum

Using the no-arbitrage identity for the simply-compounded forward rate:

$$B(t_0, t_{i-1}) - B(t_0, t_i) = E[\delta(t_{i-1}, t_i) L(t_{i-1}, t_i) B(t_0, t_i)],$$

the floating coupons telescope:

$$\sum_{i=1}^n [B(t_0, t_{i-1}) - B(t_0, t_i)] = B(t_0, t_0) - B(t_0, t_N) = 1 - B(t_0, t_N).$$

Substituting back, the $B(t_0, t_N)$ terms cancel:

$$P = N (1 - B(t_0, t_N)) + N B(t_0, t_N) = N.$$

Result

$$P = N = 100 \text{ Mln}$$

The IB coupon bond prices at **par** and this follows directly from the coupon being set at the fair swap rate.

Exercise 3: Asset Swap Spread

An asset swap packages a fixed-coupon bond with an interest rate swap so that the investor receives Euribor 3m plus a constant spread s instead of the fixed coupons. The spread is determined by a zero-NPV condition.

Dirty price. The bond (issued 31 March 2007, maturity 31 March 2012, coupon $c = 4.6\%$ annual, face value $N = 100$) has accrued interest $A = c \cdot N \cdot (t_0 - t_{\text{prev}})/(t_{\text{next}} - t_{\text{prev}})$, giving $P^{\text{dirty}} = P^{\text{clean}} + A$.

Risk-free PV of bond cash flows. Future coupons and principal are discounted with the bootstrapped EUR curve:

$$\text{PV}_{\text{bond}} = c \cdot N \sum_{k=1}^n B(t_0, t_k) + N \cdot B(t_0, T).$$

BPV of the floating leg. The Euribor 3m leg pays quarterly (act/360); its basis-point value is:

$$\text{BPV}_{\text{ft}} = N \sum_{j=1}^m \tau_j B(t_0, u_j), \quad \tau_j = \frac{u_j - u_{j-1}}{360}.$$

Asset swap spread. Setting the NPV of the package to zero yields:

$$s = \frac{\text{PV}_{\text{bond}} - P^{\text{dirty}}}{\text{BPV}_{\text{ft}}}$$

Results

Quantity	Value
Clean price P^{clean}	101.5000
Accrued interest A	4.0761
Dirty price P^{dirty}	105.5761
PV bond (risk-free)	106.4364
ASW spread s	22.44 bps

Table 1: Asset swap spread for issuer YY bond, 15 Feb 2008.

The positive spread of **22.44** bps indicates that the bond's risk-free present value exceeds its dirty price: the fixed coupon of 4.6% is above the prevailing Euribor curve level, so the issuer YY bond trades at a premium to the swap curve.

4. Case Study: CDS Bootstrap

This section analyzes the bootstrapping of survival probabilities and default intensities from Credit Default Swap (CDS) spreads on the 15th of February 2008 at 10:45 C.E.T.. The obligor evaluated has a recovery rate of 40% and market spreads for the annual CDS are 29 bps (1y), 34 bps (2y), 37 bps (3y), 39 bps (4y), 40 bps (5y), and 40 bps (7y).

1. Implementation

To perform the bootstrap, a complete set of annual spreads up to 7 years is required. The missing 6-year CDS spread was interpolated using a cubic spline.

The bootstrapping algorithm iteratively extracts piecewise constant default intensities, $\lambda(t)$, and corresponding survival probabilities by equating the Premium Leg and the Protection Leg. The implementation compares three distinct methodologies, defining $B(t_0, t_i)$ as the discount factor, $P(t_0, t_i)$ as the survival probability, π as the recovery rate, and $\Delta t_i = t_i - t_{i-1}$:

- **Approximated Method:** Computes the protection leg by neglecting the accrual premium paid between the last payment date and the default time.

– **First step** ($i = 1$):

$$s_1 \cdot \Delta t_1 \cdot B(t_0, t_1) \cdot P(t_0, t_1) = (1 - \pi) \cdot B(t_0, t_1) \cdot [1 - P(t_0, t_1)]$$

– **Generic step** ($i > 1$):

$$s_i \sum_{j=1}^i \Delta t_j \cdot B(t_0, t_j) \cdot P(t_0, t_j) = \sum_{j=1}^i (1 - \pi) \cdot B(t_0, t_j) \cdot [P(t_0, t_{j-1}) - P(t_0, t_j)]$$

- **Exact Method:** Accounts for the accrual term by assuming default occurs exactly halfway through the respective time interval.

– **First step** ($i = 1$):

$$s_1 \cdot \Delta t_1 \cdot B(t_0, t_1) \cdot P(t_0, t_1) + s_1 \cdot \frac{\Delta t_1}{2} \cdot B(t_0, t_1) \cdot [1 - P(t_0, t_1)] = (1 - \pi) \cdot B(t_0, t_1) \cdot [1 - P(t_0, t_1)]$$

– **Generic step** ($i > 1$):

$$\begin{aligned} s_i \sum_{j=1}^i \Delta t_j \cdot B(t_0, t_j) \cdot P(t_0, t_j) + s_i \sum_{j=1}^i \frac{\Delta t_j}{2} \cdot B(t_0, t_j) \cdot [P(t_0, t_{j-1}) - P(t_0, t_j)] \\ = \sum_{j=1}^i (1 - \pi) \cdot B(t_0, t_j) \cdot [P(t_0, t_{j-1}) - P(t_0, t_j)] \end{aligned}$$

- **Jarrow-Turnbull (JT) Approximation:** Uses a continuous and constant formulation where the intensity is simply proportional to the spread, calculated as $\lambda = \frac{s}{1 - \pi}$.

2. Results and Discussion

2.1 Impact of the Accrual Term

The outputs demonstrate that the difference between the Approximated and Exact methods is remarkably small. Table 2 compares the forward default intensities for both methods.

Year	Intensity (Approx)	Intensity (Exact)
1	0.004822	0.004833
2	0.006518	0.006538
3	0.007212	0.007237
4	0.007571	0.007599
5	0.007391	0.007418
6	0.006889	0.006912
7	0.006387	0.006407

Table 2: Comparison of Approximated vs. Exact Bootstrapping

The maximum absolute difference between the intensities calculated via the exact and approximate methods (`max_diff`) is 2.8038×10^{-5} , which equates to roughly 0.28 basis points. This confirms the theoretical expectation that the accrual term’s impact is negligible for standard pricing purposes.

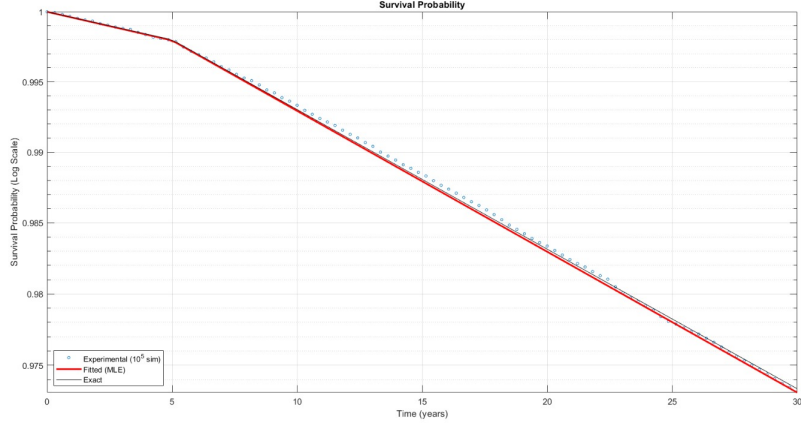
2.2 Jarrow-Turnbull Approximation

The Jarrow-Turnbull approach yields a different profile for the intensities. Because the JT method is non-recursive and calculates a constant intensity for each maturity directly from the spread, it does not extract ”forward” intensities.

Year	Intensity (JT)
1	0.004833
2	0.005667
3	0.006167
4	0.006500
5	0.006667
6	0.006703
7	0.006667

Table 3: Jarrow-Turnbull Approximation Results

As seen in Table 3, the JT intensities smoothly follow the shape of the CDS spreads (scaling them by $1/0.6$). However, this method underprices the marginal risk in the middle years compared to the exact bootstrap. For instance, the exact method isolates a forward intensity of 7.6 bps in Year 4 to account for the jump in the cumulative spread, whereas the JT method spreads this risk evenly from Year 0 to Year 4, resulting in a flatter intensity curve (6.5 bps). Consequently, while JT is computationally trivial, it fails to capture the true term structure of default risk dynamically.



5. Exercise : Credit Simulation and Default Time Estimation

This section details the simulation of default times based on a piecewise constant hazard rate and the estimation of these parameters using Maximum Likelihood Estimation (MLE).

5.1 Default Time Simulation

The survival probability of a credit entity up to time T is given by $P(t, T) = e^{-\int_t^T \lambda_s ds}$. We assume a piecewise constant hazard rate term structure where $\lambda_1 = 4$ bps for $s \leq \theta$ and $\lambda_2 = 10$ bps for $s > \theta$, with the threshold $\theta = 5$ years.

To simulate the random default time τ , we employ the **Inverse Transform Sampling** method. Since the cumulative distribution function of τ is uniform on $[0, 1]$, the survival probability evaluated at τ is also uniformly distributed: $U = P(0, \tau) \sim \mathcal{U}(0, 1)$.

By drawing a uniform random variable U , we can analytically invert the survival function to solve for τ :

- **Default before θ ($U \geq e^{-\lambda_1 \theta}$):**

$$U = e^{-\lambda_1 \tau} \implies \tau = -\frac{\ln(U)}{\lambda_1}$$

- **Default after θ ($U < e^{-\lambda_1 \theta}$):**

$$U = e^{-\lambda_1 \theta - \lambda_2 (\tau - \theta)} \implies \tau = \theta - \frac{\ln(U) + \lambda_1 \theta}{\lambda_2}$$

If a simulated default occurs after 30 years, we cap the time at $T = 30$

5.2 Maximum Likelihood Estimators and Confidence Intervals

To recover the parameters from our $M = 10^5$ simulated samples, we treat the default events as a Poisson process. Let N be the number of observed defaults in a given time

interval and T_{risk} be the total time at risk (exposure) aggregated across all simulations in that interval. The likelihood function for a constant rate λ is proportional to:

$$L(\lambda) \propto \lambda^N e^{-\lambda T_{\text{risk}}}$$

Taking the natural logarithm yields the log-likelihood function:

$$\ell(\lambda) = N \ln(\lambda) - \lambda T_{\text{risk}}$$

To find the maximum, we set the first derivative with respect to λ to zero:

$$\frac{\partial \ell}{\partial \lambda} = \frac{N}{\lambda} - T_{\text{risk}} = 0 \implies \hat{\lambda}_{\text{MLE}} = \frac{N}{T_{\text{risk}}}$$

Applying this to our piecewise framework, we calculate $\hat{\lambda}_1 = N_1/T_1$ using data exclusively up to θ , and $\hat{\lambda}_2 = N_2/T_2$ for the surviving exposure after θ .

According to the *Asymptotic Normality of Maximum Likelihood Estimators* theorem, for large sample sizes, $\hat{\lambda}$ is approximately normally distributed: $\hat{\lambda} \sim \mathcal{N}\left(\lambda, \frac{\lambda^2}{N}\right)$. Using the estimator itself to approximate the standard error, the 95% bilateral confidence interval is calculated as:

$$\text{CI}_{95\%} = \left[\hat{\lambda} - 1.96 \frac{\hat{\lambda}}{\sqrt{N}}, \hat{\lambda} + 1.96 \frac{\hat{\lambda}}{\sqrt{N}} \right]$$

5.3 Results and Plot Analysis

Applying the MLE framework to our simulated data yields the following estimates:

- **Lambda 1** (exact 4 bps): Estimated at **4.10 bps** — CI 95%: [3.54, 4.67]
- **Lambda 2** (exact 10 bps): Estimated at **10.09 bps** — CI 95%: [9.69, 10.48]

The estimation procedure is highly successful. Both theoretical values fall neatly within their 95% confidence intervals, robustly validating the underlying simulation logic and the statistical inference.

Regarding the semi-logarithmic plot of the survival probabilities, an observable behavior is that the exact theoretical curve deviates slightly from the experimental and fitted curves toward the 30-year mark. This discrepancy is a natural manifestation of Monte Carlo sample variance, rather than a logical or coding error.

Specifically, because our specific random sample yielded estimated rates ($\hat{\lambda}_1 = 4.10$ bps, $\hat{\lambda}_2 = 10.09$ bps) that are fractionally higher than the exact rates (4 bps and 10 bps), defaults occurred at a slightly faster pace in the simulation. Over a long 30-year horizon, this minor excess in the empirical hazard rate compounds exponentially, causing the simulated survival curve to drop marginally below the theoretical one at the tail end.

6. Exercise: Valuation of Monthly Cash Flows with an AAGR

1. Methodology

This section evaluates the Net Present Value (NPV) of a monthly cash flow stream subject to an Average Annual Growth Rate (AAGR) under two different initial scenarios.

To accurately price this growing annuity, the valuation algorithm applies a discrete annual step-up to the cash flows. The discounting process leverages the zero-rate curve bootstrapped in Exercise 1. For each of the 240 specific payment dates, precise zero rates were obtained via linear interpolation of the year fractions, which were then converted into exact discount factors.

The NPV is computed using the following general formula:

$$\text{NPV} = \sum_{i=1}^{240} CF_i \cdot B(t_0, t_i)$$

where CF_i represents the monthly cash flow (growing by 5% annually), and $B(t_0, t_i)$ is the discount factor interpolated for the specific payment date t_i .

2. Results

The NPV was computed under two different initial monthly cash flow scenarios.

Scenario	Initial Cash Flow	Net Present Value (NPV)
Case A	1,500 €	363683.03€
Case B	6,000 €	1454732.11€

3. Financial Considerations

The results highlight two key points:

- **Proportionality:** The initial cash flow in Case B (6,000 €) is exactly four times the cash flow in Case A (1,500 €). Because the NPV calculation is a linear mathematical operator, the final NPV for Case B scales by the exact same factor, resulting in precisely four times the NPV of Case A. This confirms the internal consistency of the pricing model.
- **Growth vs Discounting Dynamics:** Typically, cash flows received far in the future have a very low present value due to the exponential decay of discount factors over a 20-year horizon. However, the 5% annual growth rate strongly counteracts this effect. By continuously compounding the nominal value of future payments, the AAGR ensures that the cash flows received in the later years still hold significant weight in the final NPV calculation.

7. Arbitrage

The synthetic repo strategy consists of selling an EUA forward with maturity T , borrowing S_t at time t , and using the borrowed funds to buy the EUA spot. At maturity, the trader delivers the EUA to settle the forward, receives $F_{t,T}$, and repays the loan with interest. If the forward price satisfies $F_{t,T} > S_t e^{R(t,T)(T-t)}$, the trader locks in a risk-free profit, creating an arbitrage opportunity. However, the arbitrage does not disappear because the financing rate includes a credit spread: market participants cannot borrow at the risk-free rate. Since EUA positions cannot easily be financed in the repo market, funding costs remain higher and the spread persists.